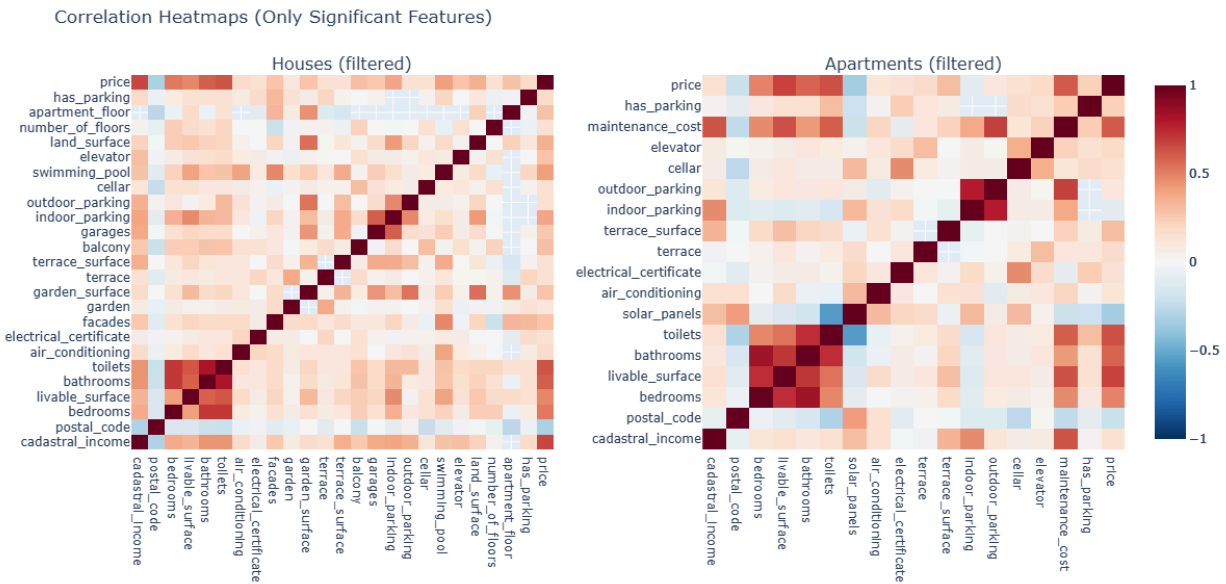




The heatmaps illustrate the correlation structure between selected variables and property prices for houses and apartments separately.

To improve interpretability, only statistically significant features were retained based on both Pearson correlation ( $r$  threshold) and  $p$ -value filtering, ensuring that the observed relationships are not due to random variation.

The results show that key variables such as livable surface, number of bathrooms, and number of toilets exhibit strong positive correlations with price in both property types. However, the intensity and patterns of correlation differ between houses and apartments, indicating distinct pricing structures.

In particular, apartments show a stronger clustering of correlated variables (e.g. surface, bedrooms, bathrooms), whereas houses display a more dispersed correlation structure, suggesting greater variability and additional influencing factors.

Overall, the heatmaps confirm that while some structural features consistently drive price, the relationship between variables and price differs depending on the property type.



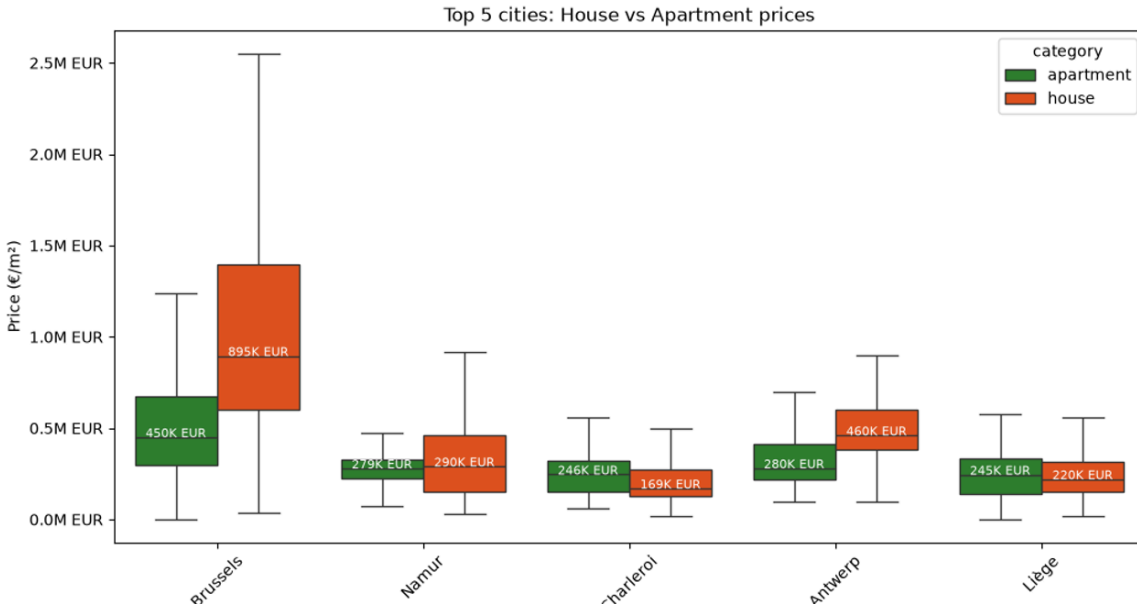
The chart presents the number of available properties in the five largest cities, segmented into apartments and houses.

Brussels clearly dominates the market, with a significantly higher number of listings compared to the other cities, particularly in the apartment segment.

Among the remaining cities, Charleroi and Namur exhibit higher availability of properties than Antwerp and Liège, despite Antwerp and Liège being larger cities in terms of population.

This suggests that the volume of listings does not necessarily follow city size and may instead reflect differences in market dynamics, housing stock, or supply activity at the time of data collection (15 June 2026).

Additionally, Charleroi shows a relatively higher proportion of houses compared to apartments, whereas Brussels is dominated by apartments, highlighting structural differences in the housing markets across cities.



The boxplot illustrates the distribution of total property prices across the top five cities, distinguishing between apartments and houses.

Brussels exhibits the largest difference in median prices between apartments and houses, with houses being significantly more expensive. This reflects the broader price range and higher variability of house prices in the capital.

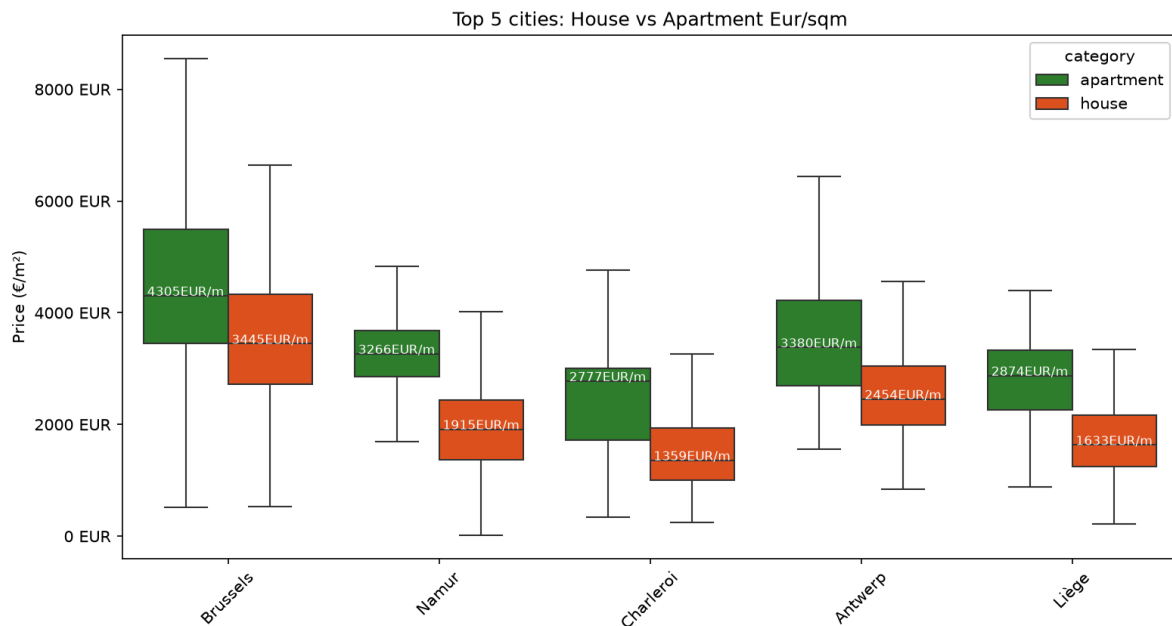
In contrast, cities such as Liège show much smaller differences between the median prices of apartments and houses, suggesting a more homogeneous pricing structure between property types.

Interestingly, in Namur and Charleroi, the median price of houses is comparable to or even lower than that of apartments. In Charleroi in particular, the distribution of house prices appears lower overall than that of apartments, indicating a distinct market dynamic where houses may not command a price premium.

Antwerp behaves more similarly to Brussels, with houses generally more expensive than apartments, although the gap is less pronounced.

These results suggest that the relationship between property type and price varies significantly across cities, reflecting differences in urban density, housing supply, and market demand.

These results suggest that the relationship between property type and price varies significantly across cities, reflecting differences in urban density, housing supply, and market demand.



The second boxplot presents the distribution of property prices per square meter across the top five cities, distinguishing between apartments and houses.

Unlike the first boxplot (total price), this representation removes the effect of property size, allowing a normalized comparison across property types and locations.

The results reveal that, in all cities, apartments consistently have a higher median price per square meter than houses. This indicates that apartments are generally more expensive relative to their size.

The difference is particularly pronounced in Charleroi, where the gap between apartments and houses is the largest. This suggests that apartments in Charleroi command a relatively higher price per unit area, despite houses having lower total prices.

Overall, while houses may appear more expensive in absolute terms, they are typically cheaper than apartments when considering price per square meter. This highlights the importance of using normalized measures to better understand market dynamics.